Band-1 Computational Experiment: Dedekind numbers — finite exact ledger M(0)..M(5)

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Honesty stamp. This note does **not** claim that Dedekind numbers are "solved," does **not** compute $M(9)$ (open), and does **not** stamp Clay-closed or a finished Theory of Everything. It box-certifies the classical small values $M(0) \dots M(5)$ by exhaustive enumeration of monotone Boolean functions, and records published values $M(6) \dots M(8)$ as **literature only**.

Abstract

The Dedekind number $M(n)$ (OEIS A000372) counts monotone Boolean functions $\{0,1\}^n \to \{0,1\}$, equivalently the number of downsets (or antichains) of the Boolean lattice B_n. Tonight's Band-1 finite ledger:

1. Implements a pure-Python backtracking enumerator ordered by popcount.
2. Independently recomputes $M(0), \dots, M(5)$ on this Linux box.
3. Checks exact equality against the standing literature table.
4. Ledgers $M(6), M(7)$ (classical) and $M(8)$ (2023 independent computations) without re-deriving them here.
5. Leaves $M(9)$ explicitly **OPEN**.

Bound / value move this run: none beyond methods validation of classical small n.

Problem statement

n	Standing $M(n)$	Source class
0	2	trivial
1	3	classical
2	6	classical
3	20	classical
4	168	classical
5	7581	Ward / OEIS
6	7828354	Church / Wiedemann
7	2414682040998	Wiedemann 1991
8	56130437209370320359966	Jäkel 2023; Van Hirtum et al. 2023

n	Standing $M(n)$	Source class
9	unknown	**OPEN**

Method

Monotone Boolean functions are enumerated by DFS over bitmasks $0 \dots 2^n - 1$ ordered by popcount. When visiting mask m, if any covered lower neighbor (clear one set bit) is already assigned 0, then $f(m)$ is forced to 0; otherwise both assignments 0 and 1 are tried. This enforces downward closure and counts each monotone function once.

Implementation: `/workspace/experiments/band1/dedekind/dedekind_finite_ledger.py` (`math`-free; stdlib only).

Method limits

- Exhaustive certification stops at $n = 5$ in this run ($M(5) = 7581$; wall ≈ 0.015 s).
- $M(6)$ is in principle enumerable ($\approx 7.8 \cdot 10^6$) but was **not** recomputed tonight — literature ledger only.
- $M(7), M(8)$ are far beyond tonight's single-unit budget; $M(8)$ rests on 2023 computer-assisted publications.
- No claim about asymptotics $\log_2 M(n) \sim \binom{n}{\lfloor n/2 \rfloor}$.

Results (numbers!)

Box-certified (exact match)

n	Box $M(n)$	Literature	Match	Wall (s)
0	2	2	yes	$< 10^{-4}$
1	3	3	yes	$< 10^{-4}$
2	6	6	yes	$< 10^{-4}$
3	20	20	yes	$< 10^{-4}$
4	168	168	yes	≈ 0.0002
5	7581	7581	yes	≈ 0.015

All six values match. Total enumeration wall ≈ 0.015 s on Python 3.13 / Linux box.

Literature-only (not recomputed)

n	$M(n)$	Note
6	7828354	Classical; not re-run tonight
7	2414682040998	Wiedemann 1991
8	56130437209370320359966	2023 dual independent computations
9	—	**OPEN**

Limitations

- Finite exact check of classical small Dedekind numbers only.
- Does not improve any published $M(n)$.
- Does not attack $M(9)$.
- No Clay / ToE / "Dedekind solved" language.

Next experiment

1. Recompute $M(6)$ on-box as a second methods certificate (expected wall seconds–minutes in optimized Python / Cython).
2. Import and spot-check published machine-checkable certificates for $M(8)$ if released in replayable form.
3. Parallel Band-1: continue Hilbert 16 / quadratic vector-field effective bounds (tonight item 6).

Conclusion

Band-1 Dedekind finite ledger complete as **hygiene / exact verification**:

$$M(0), \ldots, M(5) = 2, 3, 6, 20, 168, 7581$$

independently reproduced; $M(6) \ldots M(8)$ ledgered from literature; $M(9)$ left open.

No new Dedekind value. No Clay-closed claim. No finished-ToE stamp. GATE-04 collection remains LOCKED.

References (selected)

- OEIS A000372 (Dedekind numbers).
- R. Dedekind (1897): Über Zerlegungen von Zahlen…
- D. Wiedemann (1991): A computation of the eighth Dedekind number (note: indexing conventions vary in older literature; we follow OEIS A000372).
- C. Jäkel (2023); L. Van Hirtum et al. (2023): independent computations of $M(8)$.

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